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Building a Quantum Simulator in Rust

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Abstract

This paper investigates different approaches to implementing quantum circuit statevector simulators and stabiliser techniques. This was done by developing a statevector simulator from scratch alongside a secondary stabiliser simulator and evaluating them through benchmarking and testing. The implementation operates directly on pairs of amplitudes rather than constructing full-system matrices. It also utilises fused multiply-add and SIMD instructions. Benchmarks showed that the single-gate and controlled two-qubit gate pair-based kernels were respectively 44,391.94 and 44,391.94 times faster than the full-system matrix approach at 13 qubits, while using 1.34 and 2.42 gigabytes less memory, and the SIMD kernel gave a further 158-225% speedup over the portable implementation. The stabiliser was 14668.68 times faster than the statevector when running the same circuit 19-qubit circuit.

1 Introduction

Quantum computing is an emerging technology that leverages principles of quantum mechanics at the hardware level to create computers that are capable of performing computations in a fundamentally different way to classical machines. For specific classes of problems this ‘quantum advantage’ can be utilised by carefully designed algorithms to achieve non-trivial improvements in computational complexity [1].

However, the field is in the Noisy Intermediate Scale Quantum (NISQ) era, characterized by processors that are constrained in size (number of qubits) and lack fault tolerance. In this context, a lack of fault tolerance means that processors are not yet stable enough to correct the errors introduced by cumulative external noise [2]. Beyond these technical limitations, the hardware remains a scarce resource; it is extremely expensive to build, maintain and run a quantum computer. While cloud-based providers, such as Microsoft Azure Quantum and Amazon Braket, have improved access to quantum computing by offering Quantum-as-a-Service (QaaS), access to high-performance chips can involve long wait times and significant usage fees, which can pose an obstacle to development [3]. Due to these constraints, the ability to simulate quantum circuits on classical hardware currently remains the most practical way to develop and test quantum algorithms. Quantum circuit simulation involves evaluating the effect of a series of logic gates mathematically on a representation of a quantum state. This approach provides an environment free of the typical environmental noise inherent to current quantum hardware and is far more cost effective [4].

However, simulation comes with its own set of limitations. The core issue is the memory cost of describing a quantum state classically. The base units of quantum information are qubits (quantum bits) and the direct conventional approach is known as statevector simulation in which to fully describe a quantum system of n qubits requires tracking 2^n individual complex amplitudes. Even on a supercomputer such as Summit [5], we can only conduct algorithmic simulations of a quantum circuit up to 47 qubits, which requires 2.8 petabytes of memory [4]. Mitigating this scaling is an active area of research with many approaches that trade exactness for tractability, but in the end this limitation is not escapable, as it is an inherent consequence of mapping quantum information onto classical hardware.

To escape this exponential scaling, alternative techniques sacrifice flexibility or exactness to make high-qubit simulation feasible. For example the stabiliser formalism allows certain classes of quantum circuit to be simulated efficiently, but cannot handle arbitrary algorithms. Because of this, statevector simulation remains the primary approach when flexibility is required making its computational cost a key bottleneck.

To explore how this bottleneck can be mitigated, this paper aims to investigate the implementation of quantum circuit simulators, with a primary focus on statevector

simulation techniques alongside exploring stabiliser simulation. For this purpose a simulator library is developed from scratch with support for single-qubit gates, controlled two-qubit gates and measurement. Different approaches for applying quantum operations to a quantum state are investigated and evaluated. The paper also explores the implementation and effects of hardware acceleration on execution time.

The specific objectives are:

- Implement a functional quantum circuit simulator using a statevector representation.
- Implement common single-qubit gates, controlled two-qubit gates and measurement.
- Implement multiple high-level approaches to computing the application of quantum operations including full-system matrix and direct pair multiplication.
- Implement multiple low-level approaches to computing the application of quantum operations utilising hardware acceleration.
- Implement a secondary stabiliser simulator.
- Evaluate the computational characteristics of approaches.

Together, these objectives allow the examination of the computational cost of statevector simulation at both the algorithmic and implementation level as well an evaluation against stabiliser techniques. The following chapters will go on to discuss the underlying theory, design and implementation of the simulator, the methods used to evaluate its performance and the results of these experiments.

2 Background

This section will introduce the basics of quantum computing using statevectors and some core principles behind some of the techniques used before introducing an alternative approach to simulation. The material presented in the first four subsections is a brief introduction to the quantum computing foundations required by this project and follows Nielsen and Chuang [6].

2.1 Qubits

A qubit can be considered the quantum analogue of a bit. Unlike a classical bit which has a value of either one or zero, a qubit is capable of existing as a linear combination of both values. More precisely a linear combination of two basis states, $|0\rangle$ and $|1\rangle$. However, when measured this qubit will collapse into one of the two basis states probabilistically. The definition of a qubit is show below:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where α and β are complex numbers whose square moduli give the probability of each outcome during collapse. For a valid qubit the total probability, or the probability of there being an outcome at all, must be one. Which gives rise to the normalisation condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

2.2 Gates

Quantum gates or operators as they are also known, are operations that are applied to qubits and form the building blocks of quantum circuits. A gate is represented by a matrix and applied using matrix-vector multiplication. This can be visualised in detail by using vector notation:

$$|\psi'\rangle = U|\psi\rangle = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{bmatrix}$$

where $|\psi'\rangle$ is the resulting changed qubit, U is an arbitrary gate matrix, a , b , c and d are the matrix coefficients and α and β are still the probability amplitudes.

All gate matrices (aside from measurement gates) are unitary, meaning they preserve the normalisation of the state and can be reversed by simply applying them again.

Common single-qubit gates include the Pauli X , Y and Z gates as well as the Hadamard gate H :

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

2.3 Statevectors

So far a single qubit has been used however, there is not much computation that can be done with only a single qubit. Quantum algorithms typically run on many-qubit

systems, so while $|\psi\rangle$ represented a qubit it was actually a single qubit statevector. A statevector represents a quantum state of n -qubits by using 2^n complex amplitudes, and when generalising across n qubits it is conventionally written in the form:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle$$

where $|i\rangle$ is the i th basis state and α_i is its associated probability amplitude. Basis states are typically numbered in binary notation rather than decimal i.e $|010\rangle$. This set of basis states are known as the computational basis and measuring a system always results in one of them. So for example, a system can be said to be in the computational basis state $|010\rangle$ when all other basis states have amplitudes equal to zero. Simulations conventionally begin in the all-zero state $|0\dots 0\rangle$. The normalisation condition can then be generalised to:

$$\sum_{i=0}^{2^n-1} |\alpha_i|^2 = 1$$

To represent a statevector computationally, a fixed ordering of the basis states is assigned, either big or little endian. This project uses big-endian ordering where qubit zero is the leftmost bit of the binary label and therefore the most significant. This allows the state to be represented in a more computation friendly format as a column vector:

$$|\psi\rangle = \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{2^n-1} \end{bmatrix} \in \mathbb{C}^{2^n}$$

As a consequence the space complexity for a statevector must be $O(2^n)$.

2.4 Multi-qubit gates

Just as with single-qubit gates a gate that acts on multiple qubits will be a matrix. The number of qubits it operates on has the following relationship with its dimensions, a k qubit gate will be represented by a $2^k \times 2^k$ matrix. A common two-qubit gate is the SWAP gate:

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

Which has the property of swapping the amplitudes of two qubits unconditionally.

Another key type of gate is the controlled gate, this is what allows branching and more complex behaviour in quantum circuits. Common controlled two-qubit gates include the controlled Pauli matrices:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad \text{CY} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{bmatrix}, \quad \text{CZ} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix},$$

The way these gates function is that they ‘take’ two qubits, a control and a target qubit. If the control qubit is one then they apply the single-qubit pauli gate to the target qubit. However this perhaps overstates the simplicity when one takes into account the fact that in a quantum circuit the control qubit can be in a superposition of both one and zero simultaneously.

2.5 Kronecker products

In section 2.2 it was stated that to apply a gate to a qubit matrix-vector multiplication is used. But once the state is made up of multiple qubits, applying valid matrix-vector multiplication would fail. Since the operation requires that the number of columns in the matrix are equal to the number of rows in the column vector. So to apply a gate matrix to a n -qubit statevector, the matrix must be expanded to a full-system matrix with 2^n columns. This is done by using the kronecker product.

The Kronecker product, denoted by $\otimes$, combines two smaller matrices to produce a larger matrix, where every element in the first matrix is multiplied by the second matrix. If matrix A has dimensions $a \times b$ and matrix B has dimensions $c \times d$ then $A \otimes B$ produces a matrix with dimensions $ac \times bd$.

Therefore it can be used to expand a gate matrix to a full-system matrix that can be applied to a quantum systems statevector. For a single-qubit gate this can be described as:

$$\tilde{U}_t = I^{\otimes t} \otimes U \otimes I^{(n-t-1)}$$

Where $\tilde{U}_t$ is the full-system matrix of a single-qubit gate with a target qubit index t and $I^{\otimes t}$ denotes the Kronecker product of t identity matrices. An identity matrix produces no effect on the statevector which is why the gate can be expanded this way without distorting the operation. For controlled two-qubit gates there exists a more complex formula:

$$\widetilde{\text{CU}}_{c,t} = P_0 \otimes I + P_1 \otimes U$$

Where $\widetilde{\text{CU}}_{c,t}$ is the full-system matrix of a controlled two-qubit gate, and the projectors P_0 and P_1 are defined as:

$$P_0 = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad P_1 = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $I|\psi\rangle = |\psi\rangle$ leaving the state unchanged, $P_0 \otimes I$ represents the branch where the controlled gate is not triggered, P_0 projects the identity matrix onto the corresponding amplitudes. By the same token, P_1 maps the operation U onto the amplitudes where the control qubit is activated. Notably this requires two full-system matrices to be constructed and then combined before being applied.

It is evident that gate application on many-qubit statevectors has an enormous computational cost. With the full-system matrices having a space complexity of $O(4^n)$ and controlled two-qubit gates requiring two full-system matrices. In addition while these operations are being executed the statevector still needs to remain in memory with its own space complexity of $O(2^n)$.

This full-system matrix approach is therefore used as the baseline implementation for evaluating alternative gate-application methods that are developed in this paper. Its performance provides a reference against which the computational and memory benefits of other strategies can be measured.

2.6 Direct pair multiplication (DPM)

Rather than attempting to apply a full-system matrix to the entire statevector, there is an approach that limits a simulation's space complexity to that of the statevector's while also improving execution time with workloads much more suited to modern processors.

This approach decomposes the 2^n -element statevector into 2^{n-1} disjoint, two-dimensional subspaces. Specifically, each subspace groups pairs of amplitudes whose binary representations differ only at the target qubits position q . For each pair the first element is at base index n_i where qubit q has a value of zero and the second element is at index $n_i + s$. s can be defined as the stride between the two pairs where $s = 2^{n-q-1}$.

Each pair can then be multiplied independently by the base 2×2 gate matrix itself, before writing the resulting vectors elements back into their original positions in the statevector:

$$\begin{pmatrix} a_{n_i} \\ a_{n_i+s} \end{pmatrix} \mapsto U \begin{pmatrix} a_{n_i} \\ a_{n_i+s} \end{pmatrix}$$

The 2^{n-1} pairs collectively contain every amplitude in the statevector once. Applying the gate to each pair therefore is mathematically equivalent to applying the full-system matrix to the entire statevector in one operation. [7]

The resulting amplitudes can be written back into the statevector while the while the two necessary original amplitudes and the four matrix coefficients are already available in memory. This adds no memory requirements other than the statevectors $O(2^n)$ space complexity during gate application.

Therefore DPM removes the overhead of applying gates but the statevector itself still holds 2^n amplitudes so memory grows exponentially with the number of qubits regardless of how gates are applied. Reducing this requires a more compact state representation which, without reducing accuracy, is only possible for a restricted class of circuit.

2.7 Gottesman-Knill theorem

The Gottesman-Knill theorem [8] states that a quantum circuit consisting only of Clifford gates can be simulated efficiently on a classical computer when it begins in a computational basis state and ends with measurements in the computational basis.

Clifford gates are any gate that transform a Pauli operator into another Pauli operator when applied to it. Any Clifford gate can be built from a sequence of Hadamard, S and CNOT gates and the Pauli operators X , Y and Z are Clifford gates themselves.

Circuits composed entirely of Clifford gates are known as stabiliser circuits, since the states they produce can always be described by the Pauli operators that stabilise them. A Pauli operator on n qubits consists of n single-qubit factors, one per qubit, where each factor is I , X , Y or Z , together with an overall sign. For example XIZ would apply X to the first qubit and Z to the third while leaving the second. A Pauli operator P is said to stabilise a state $|\psi\rangle$ if:

$$P|\psi\rangle = |\psi\rangle$$

where applying it leaves the state unchanged. A stabiliser state of n qubits can be described as list of n Pauli operators. Each operator is stored as n bits marking the qubits with an X factor and n bits marking the qubits with a Z factor (Y is marked as both) plus a single bit for the sign. The whole state can thus only requires $O(n^2)$ bits rather than a statevectors $O(2^n)$ amplitudes.

3 Design & Implementation

If I had to try and put my finger on my biggest issues with this, it's that I don't seem to be able to find a structure that makes sense or feels natural. I'm also struggling to put clear boundaries between higher and lower level aspects.

3.1 Development environment

Rust was chosen as the primary development language for its performance and safety guarantees while still allowing unsafe, direct memory management if necessary. Its zero-cost abstractions allow higher level intuitive interfaces to be used without causing additional runtime overhead.

Git was used for version control throughout so that branches could be used to develop experimental features in isolation from the main implementation. Additionally this allowed individual experiments and benchmarks to be associated with specific commits for reproducibility.

GitHub was used to host the source code and provide continuous integration through GitHub Actions. Automated unit tests were run on pushes to the default branch, on Windows and Ubuntu environments, providing correctness checks across platforms, ensuring that changes did not introduce subtle logic bugs and or unnacceptably compromise accuracy.

3.2 Library Architecture & Implementation

A software library was selected as the project format. This way the simulation can be exposed through a programmatic interface rather than being directly coupled to a particular application or user interface. In future it could be extended with a simulator application that utilises the library while the functionality remains independently accessible.

The library can be organised into three principle layers: the circuit representation, simulation state objects and kernels. The circuit representation describes the operations to be executed, simulation objects maintain simulation state and coordinate execution and kernels implement the mathematical transformations required by operations.

3.2.1 Circuit Representation

This project separates the description of a quantum circuit from the statevector object used to simulate its execution. Quantum operations are represented as instructions defining the operations themselves and contain parameters pertaining to non-implementation specific information. For example rotation gate angles and qubit targeting indices. This results in a clean interface that allows different simulation objects to use different state representations while sharing the same circuit representation, as shown by the statevector and stabiliser implementations.

The core component of this interface is the Instruction enum which provides variants describing different quantum operations, mainly gates. Each variant defines the parameters that may accompany it explicitly, creating a fixed structure for required inputs. The use of an enum also offers a common type the operations can be stored and processed through, making it straightforward to express an entire circuit as an ordered collection of instructions. That ability naturally lends itself to future extensions such as circuit level optimisation as the sequence could be analysed and transformed before execution.

3.2.2 Statevector

A simulation starts by initialising a simulation object such as a statevector, with the desired number of qubits. That statevector then owns the quantum state being simulated using a heap-backed buffer of amplitudes and supplies the interface through which operations can be applied. To progress the simulation an instruction or a circuit as a collection of instructions must be passed to the statevector for execution.

The statevector then acts as a validation and dispatch layer above the operations themselves during execution, while owning the amplitude buffer. By first matching on the instruction the statevector can extract the parameters to perform the necessary validation checks. Validation within the instructions themselves was considered, but checks such as target qubit bounds depend on the state itself. Keeping it in the statevector avoids splitting related checks between the instructions and the statevector object.

After validation, the statevector translates the qubit indices into statevector strides for DPM, its primary simulation strategy, where a stride represents the distance between corresponding amplitudes in the amplitude buffer. This is determined by the position of the target qubit within the chosen big-endian structure. Despite appearing to be a low-level implementation detail, generating the strides in this layer, decouples the kernels that will receive them from the memory layout of the buffer.

The statevector also encapsulates a performance configuration that specifies the hardware features to be utilised during execution. This configuration can be resolved automatically at runtime by detecting the host's CPU capabilities or it can be supplied explicitly to request a specific configuration, which is then validated against available host features. This validation is necessary because executing a kernel that uses unsupported hardware features can result in undefined behaviour.

Qubit measurement is handled directly by the statevector and follows a different execution path from instructions. Unlike unitary operations such as gate application, it returns a value and the circuit representation has no mechanism for collecting results, so measurement does not fit into it naturally. A measurement first calculates the marginal probability of the target qubits outcomes and then samples the outcome by comparing a random number against the probability of it being zero. It then

collapses the state by setting amplitudes representative of the false outcome to zero renormalising the remaining amplitudes. Both the probability calculation and the collapse step traverse the amplitude buffer by target stride in the same manner as the DPM kernels. They are currently both implemented directly in the statevector with no kernel dispatch but could be separated into kernels in a future implementation as both could benefit steps could benefit from hardware acceleration.

3.2.3 Kernels

The kernels are separated from the main statevector implementation in a submodule containing four implementations: portable, FMA and AVX. Each performs equivalent operations using different levels of hardware-specific optimisation. The portable implementation provides the DPM kernels and, for reference, the full-system matrix kernels, the FMA implementation provides DPM kernels that use fused multiply-add operations for arithmetic and the AVX implementation provides SIMD single-qubit kernels that adapt DPM but to process multiple amplitude pairs simultaneously. In order to simplify dispatch, all kernels share the same function signatures: a mutable slice of the amplitude buffer, a four element matrix and a target stride calculated by the statevector (controlled two-qubit kernels also require a control stride).

3.2.3.1 Portable DPM kernels

This is the baseline DPM implementation designed to be the baseline strategy executed with no hardware feature requirements. It can conceptually be split into two categories, functions that traverse the amplitudes and identify pairs of amplitudes and a function that, given these pairs and a matrix, performs the matrix-vector multiplication.

The single-qubit gate kernel consists of an outer loop that walks through the statevector in blocks. Each block contains two halves, each the length of a target stride, corresponding to a different value of the target qubit. An inner loop then traverses the first half of each block pairing each amplitude with the corresponding amplitude one target stride ahead.

This is especially giving me great difficulty to explain, maybe I can come up with a diagram:

The controlled two-qubit gate kernel adds one further condition: only amplitudes pairs in which the value of the control qubit is one can be updated. The traversal depends on which stride is larger. If the control stride is larger, the first loop steps through the amplitudes in blocks the size of two control strides. Then within the second half of the control block where the control qubit is one, the single-qubit gate traversal can be applied unchanged. If the target stride is larger then the outer loop traverses the amplitudes in blocks the length of two target strides, before a second loop walks through each half where the target qubit is zero and a final inner loop selects amplitudes in sub-blocks where the control is one and pairs each amplitude with another one a target stride ahead.

Matrix application for the kernels is delegated to a dedicated helper function that is called once per amplitude pair. It copies the pairs by value into a buffer on the stack, initialises a second stack buffer to zero for the results, performs the matrix-vector multiplication and then overwrites the original amplitudes with the results. This function can be considered the most performance-critical point in the program as it runs once for every updated pair, so keeping the pair and its results on the stack avoids any execution time penalties due to heap allocations.

3.2.3.2 FMA DPM kernels

A set of kernels with traversals algorithmically identical to the portable ones but with arithmetic changes to the matrix application. During matrix-vector application fused multiply-add functions are used. These functions only guarantee the result and not the use of hardware FMA instructions. Therefore, the matrix application function is therefore explicitly compiled with hardware FMA instructions using compiler directives. The DPM traversals must be duplicated in this implementation in order to compile them with the same directives since despite not containing FMA instructions it is required for inlining.

3.2.3.3 AVX DPM kernels

This is especially giving me great difficulty to explain, maybe I can come up with a diagram:

The AVX kernel follows the same traversal as the portable DPM kernel but processes two amplitude pairs per iteration using 256-bit SIMD vectors. Each vector holds four 16-byte double precision floating point values which is enough for two complex amplitudes stored as real and imaginary components. It is compiled with the AVX compiler directive to ensure that the processors SIMD YMM registers are used rather than the vectors being split across two 128-bit registers.

Updating two neighbouring low amplitudes together requires that both are in the first half of the same block, as in order for them to be loaded simultaneously they must be contiguous in memory, so the target stride must be at least two. A target stride of one means that every pair is interleaved in memory, so must be handled by other kernels instead. The outer loop walks the statevector in blocks of two target strides while the inner loop steps through the first half of each block in steps of two. Each iteration therefore updates two consecutive low amplitudes along with their respective partners one target stride ahead.

The matrix is prepared once before the traversal begins, each of the eight values that make up the four complex matrix coefficients are copied across every lane of their own vectors. This is so that a single vector multiplication can apply the same coefficient to both amplitudes held in a register. As the matrix values are invariant across the entire gate application these vectors are created once per kernel call and passed to the pair routine rather than rebuilding them for every pair. An additional

vector containing alternating signs used during complex multiplication is prepared in advance as well.

The main role of the kernel is to multiply a vector of amplitudes by one complex matrix coefficient. This is done by multiplying the amplitude vector by the real coefficient vector, and by the imaginary coefficient vector, both products are then stored separately in their own vectors. To facilitate complex multiplication the lanes of the imaginary products are swapped with their neighbouring lane, before adding the two product vectors and multiplying the result by the sign vector.

The matrix-vector product can then be constructed directly from this operation. The two low amplitudes are packed into a vector and the two high amplitudes into another. The updated low amplitudes are the sum of the top left coefficient applied to the low vector and the top right coefficients applied to the high vector. The updated high amplitudes follow the same process but with the bottom left and right coefficients respectively. Then the vectors are unpacked and written back to the amplitude buffer.

3.2.4 Full-system matrix kernels

Gate application in the full-system matrix kernels acts exactly as described in the Kronecker products background section. They exist as the baseline against which the DPM kernels are measured. They are retained in the portable module but marked as deprecated so that any use outside benchmarks and tests is flagged by the compiler. As the full-system matrix is defined in terms of qubits rather than strides, the kernels first recover the number of qubits using the logarithm of the amplitude count and convert each stride back into a qubit index.

To apply a single-qubit gate the kernel builds the full-system matrix by combining matrix factors using kronecker products across the qubits from first to last. At each position the factor is either the identity matrix, or the gate matrix if the position is the target qubit. The product itself is calculated by a helper that iterates over every factor to apply the next one to the intermediate result. The resulting full-system matrix is then multiplied by the statevector into an output buffer the same length as the state. Only once every amplitude has been computed, the buffer is copied back over the original amplitudes, as overwriting them any earlier would corrupt the inputs for other rows.

The controlled two-qubit kernel follows the projector formulation from the background section but generalises to arbitrary control and target positions. Two full-system matrices are built with the same process as the single-qubit kernel. In the first matrix m_0 the control position receives the projector matrix p_0 , shown in section 2.5, and all other positions receive an identity matrix. This represents the branch where the gate is not triggered. In the second matrix m_1 the control position receives projector matrix p_1 , the target position receives the gate matrix and all other positions again receive the identity matrix. In the same manner as the single-qubit kernel, a helper iterates over these factors and constructs both m_0 and m_1 . Rather than adding the two

matrices into a third, the matrix-vector multiplication adds the corresponding elements of both as it applies the matrix to the amplitudes. This avoids a further allocation of the same size but it still requires both matrices to be in memory at once.

3.2.5 Stabiliser

The stabiliser simulator is a small secondary investigation alongside the statevector implementation and applies the Gottesman-Knill theorem introduced in the background chapter. It takes on the same role as the statevector in the library as a simulation object although is not split into a dispatch and kernel layer due to its low complexity.

Its state is a tableau of $2n$ rows where the first n rows are destabilizers and the remaining n rows are stabilizer generators. Each row stores an X bit and a Z bit for every qubit along with a phase bit recording its sign, giving $2n(2n + 1)$ bits in total. While not covered in the background, tracking an equal number of destabilisers provides an alternative measurement method which reduces the time complexity of measurement from x to y .

Measurement for the stabiliser state was not deemed a priority and thus is unimplemented, however destabilisers are tracked in expectation of a future implementation. Despite lacking measurement, the correctness of the stabiliser can be validated against the statevector simulator. The same circuit is run on both simulators starting from the a zero state. The statevector simulator produces the final state while the tableau supplies its n stabiliser generators. Then as the generators represent the actual Pauli gates it's possible to iterate over them and construct a circuit of instructions equivalent to the tableaux internal state. This circuit can then be run on a second zero-initialised statevector and compared to the first.

3.3 Testing, benchmarking and observability

Unit tests were used constantly throughout every step of development to ensure the behaviour of the simulator was as expected and inaccuracies were not introduced. Due to the use of floating point arithmetic a certain degree of accumulated numerical error is inevitable. However a tolerance of 10^{-14} was the maximum allowed difference during tests.

For example, using the statevector-stabiliser test described above every circuit of up to three gates on three qubits drawn from the supported gates each stabiliser generator including its sign must leave the state produced by the statevector simulator unchanged. This covers 6175 circuits and tests the phase updates directly.

Criterion.rs, a Rust port of the popular Haskell Criterion package, was used for execution time benchmarking and analysis. It iterates over benchmarks gathering samples and then provides a full statistical breakdown. It also supports direct comparison benchmarks useful for verifying the difference between different approaches and algorithms. This was most useful when it came to developing

different gate application algorithms and analysing the effects of compiler directive placement.

DHAT was used to investigate the behaviour of heap allocations. It is registered as the global allocator and wraps the default allocator so that every allocation and deallocation is recorded while profiling is active. From this it can report the total number of allocations and bytes requested alongside peak heap usage. Each allocation also records the backtrace of the call that made it, which aids when analysing what these allocations can be attributed to.

The Tracing crate was used to provide observability insights into circuit execution. Primarily useful when examining sub-routines such as the quantum Fourier transform which expands into a series of gates whose ordering depends on parameters such as qubit count. It provided quality-of-life features such as exporting trace data to Chrome Perfetto for examination.

Rust's feature gating was used extensively throughout in coordination with these features to ensure that any accommodations for benchmarking and tracing did not introduce unnecessary overhead when unactivated.

4 Results

This chapter compares the correctness and computational cost of the various strategies taken. It examines how execution time and memory usage evolve over an increasing number of qubits. In the majority of cases logarithmic graphs are preferred due to the exponential nature of the data.

4.1 Experimental configuration & Methodology

The following results were achieved using a desktop computer equipped with a Ryzen 5 1600 processor and 16 GB of RAM. The operating system was CachyOS, a Linux distribution. The benchmarks were compiled and run on the nightly-2026-05-27 Rust toolchain.

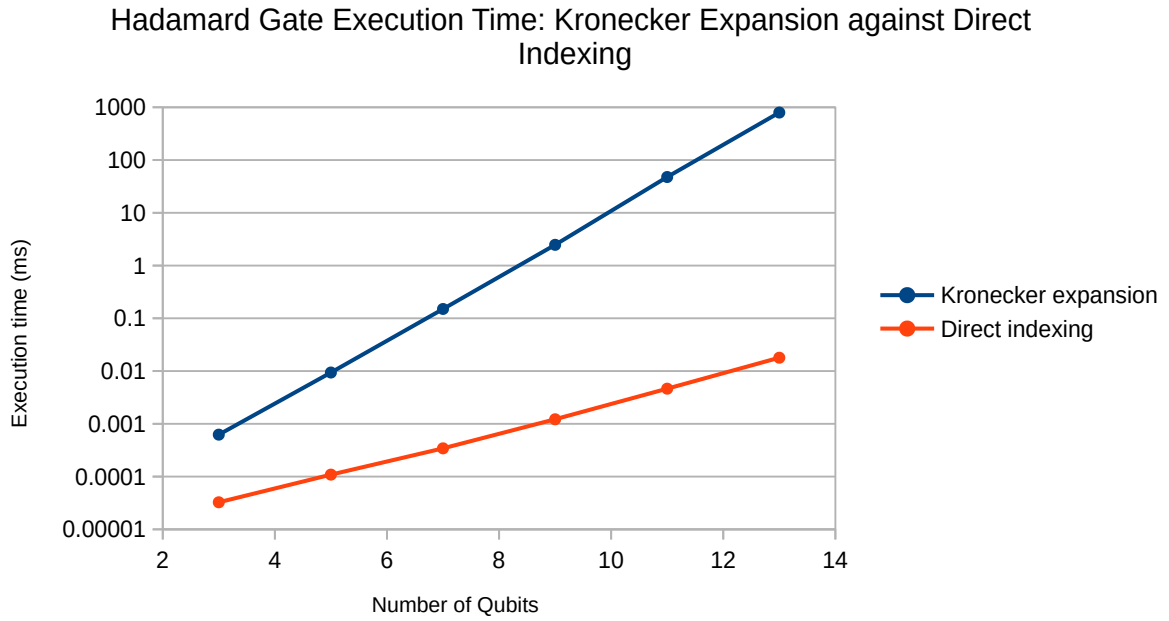
Each experiment that measured execution time was run using Criterion's benchmarking harness. Before collecting measurements, Criterion performed a three second warm-up period during which the benchmark is repeatedly executed without measurement in order to allow the system and host machines hardware to adapt to the load. It then ran benchmarks repeatedly over thousands of iterations to obtain 100 samples. It then provided a statistical analysis of said samples providing a mean, standard deviation and confidence intervals.

When comparing kernels directly against each other, amplitude buffers were constructed explicitly outside of the benchmark and kernels were called directly in benchmarks to remove any overhead that the validation and dispatch of the statevector object could introduce.

4.2 Full-system matrix against portable DPM

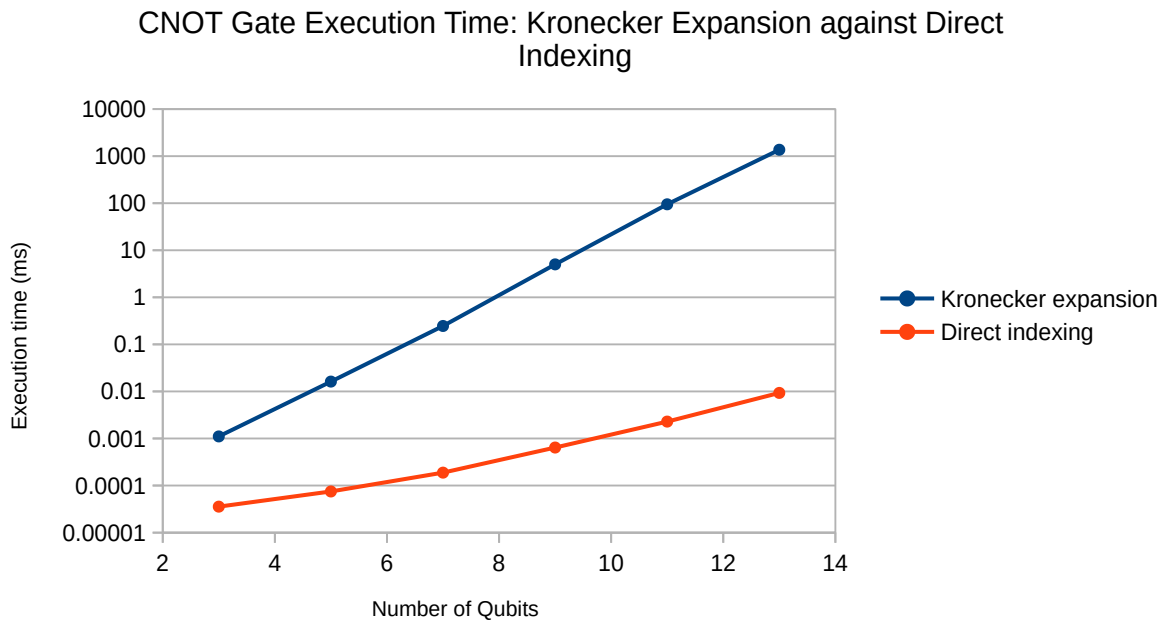
The first comparison evaluates expansion of the full-system matrix against the baseline portable DPM approach. The two strategies were benchmarked using singular Hadamard and CNOT gates over amplitude buffers containing an increasing number of qubits from $n = 3$ to $n = 13$ in increments of two. These experiments correspond to both the single-qubit and controlled two-qubit kernels. To isolate qubit count as the independent variable, the target was always the middle qubit and for CNOT gates the control qubit was always selected as zero.

4.2.1 Hadamard gate execution time



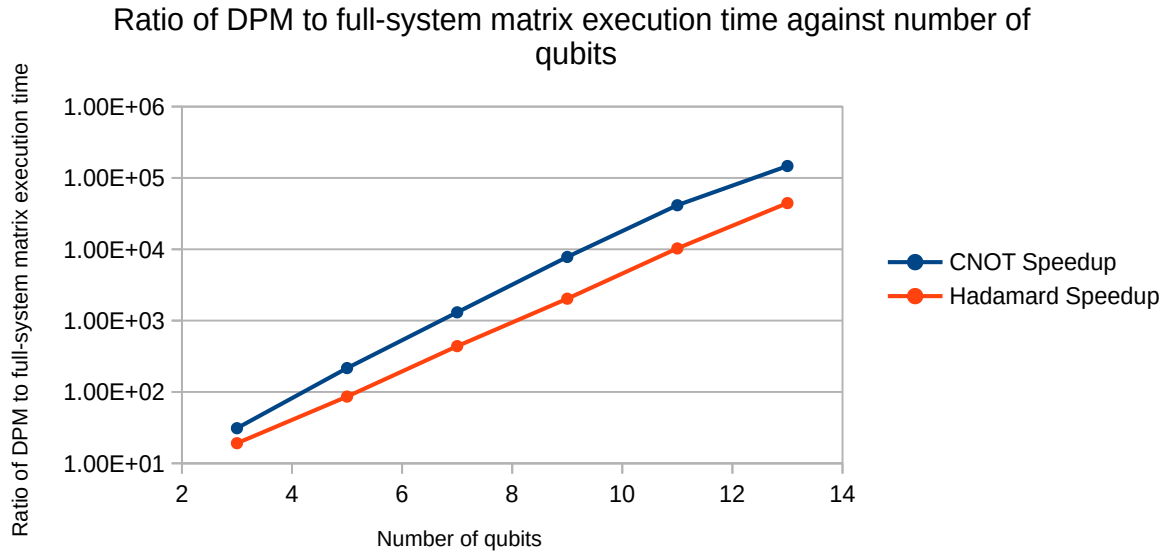
The gap between the two implementations grows exponentially with the number of qubits. At $n = 3$ the DPM kernel is 19.13 times faster than full-system expansion and at $n = 13$ it is 44,391.94 times faster.

4.2.2 CNOT gate execution time



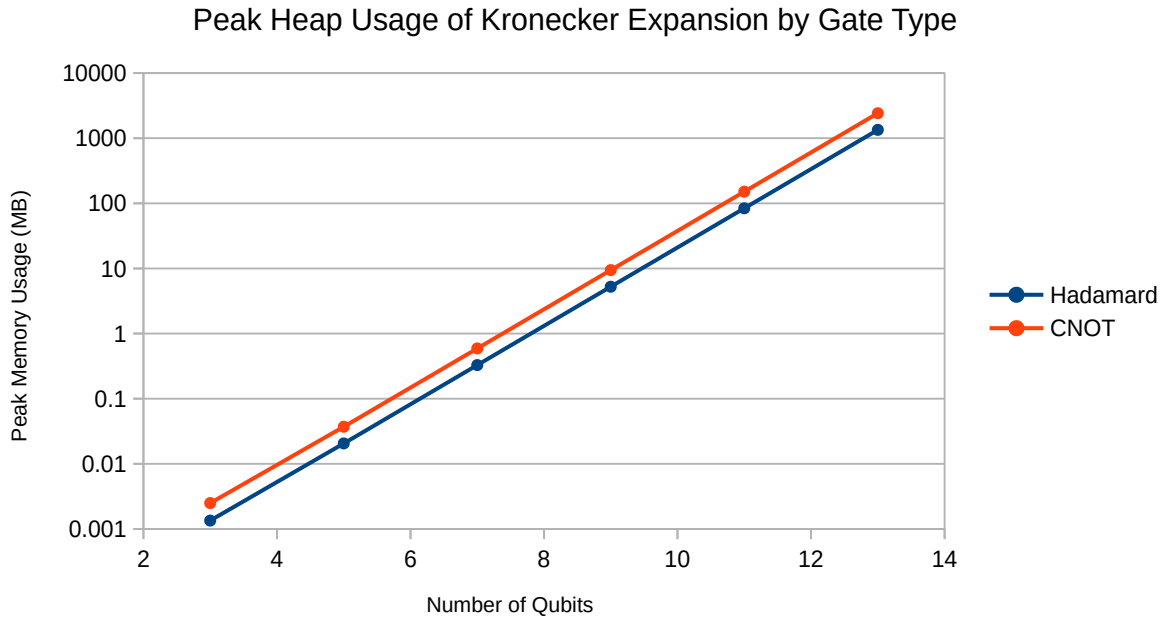
The controlled two-qubit kernels show an even larger difference. The speedup is 31.03 at $n = 3$ and 147,010.09 at $n = 13$.

4.2.3 Execution time gains



4.2.4 Hadamard and CNOT gate peak memory usage

DPM was excluded from the following graphs, since as expected, its gate applications triggered no additional allocations. Instead the difference in memory usage between the two full-system matrix expansion kernels can be examined.

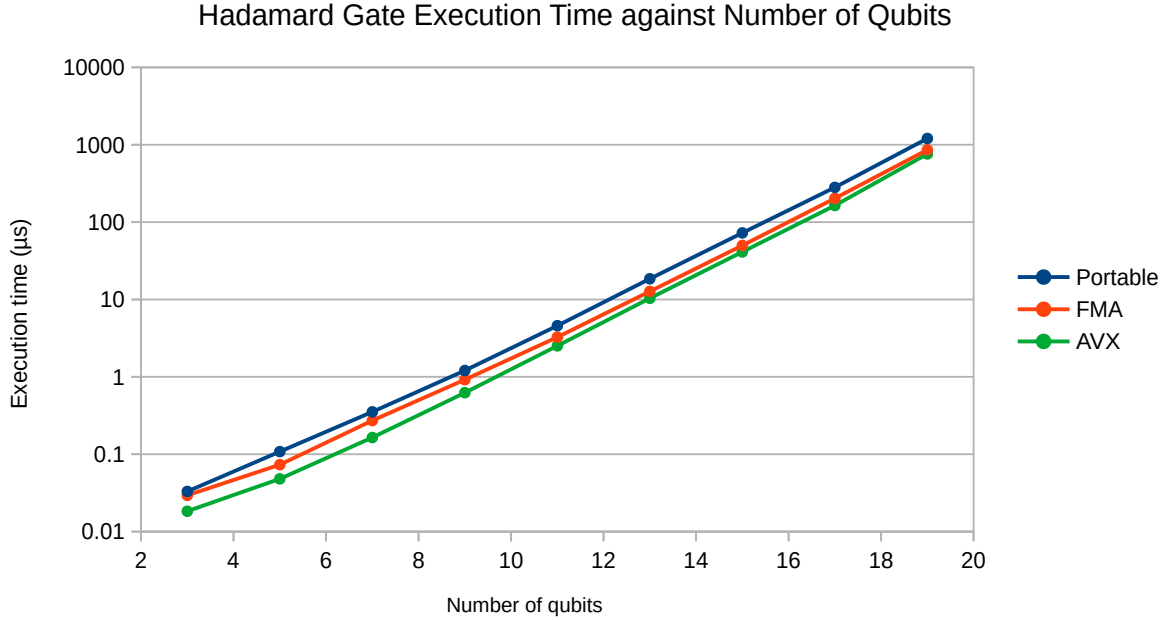


At $n = 3$ the controlled two-qubit kernel consumes 1.86 times the peak memory as the single-qubit kernel. At $n = 13$ this factor had approached 1.80. At this point the single-qubit kernel reaches a peak memory usage of approximately 1.34 GB, while the controlled two-qubit reaches 2.42 GB.

4.3 Hardware accelerated kernels

The following experiments evaluate the effect of fused multiply-add operations and SIMD instructions on the execution time of DPM statevector gate application.

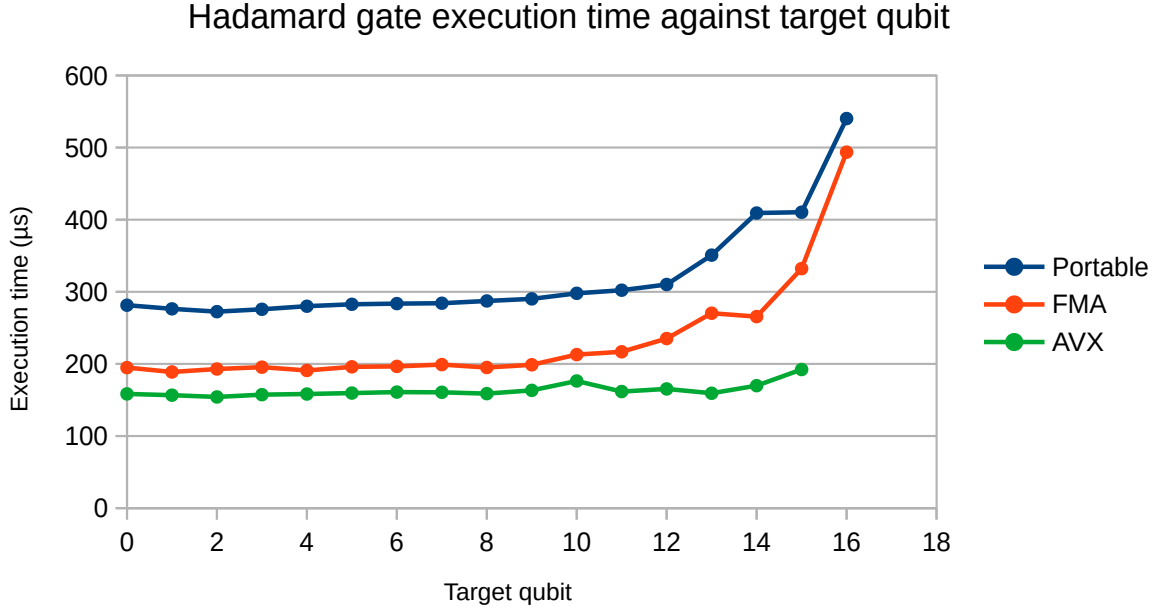
4.3.1 Hadamard gate execution time over number of qubits



All three implementations show the expected exponential increase in execution time as qubit count increases. However both hardware accelerated kernels consistently outperform the portable kernel, with a significant reduction in execution time from the AVX kernel. At $n = 3$ the speedup from the FMA and AVX kernels are 1.12 and 1.80 respectively while at $n = 19$ this has increased to 1.40 and 1.58.

4.3.2 Hadamard gate execution time over target qubit

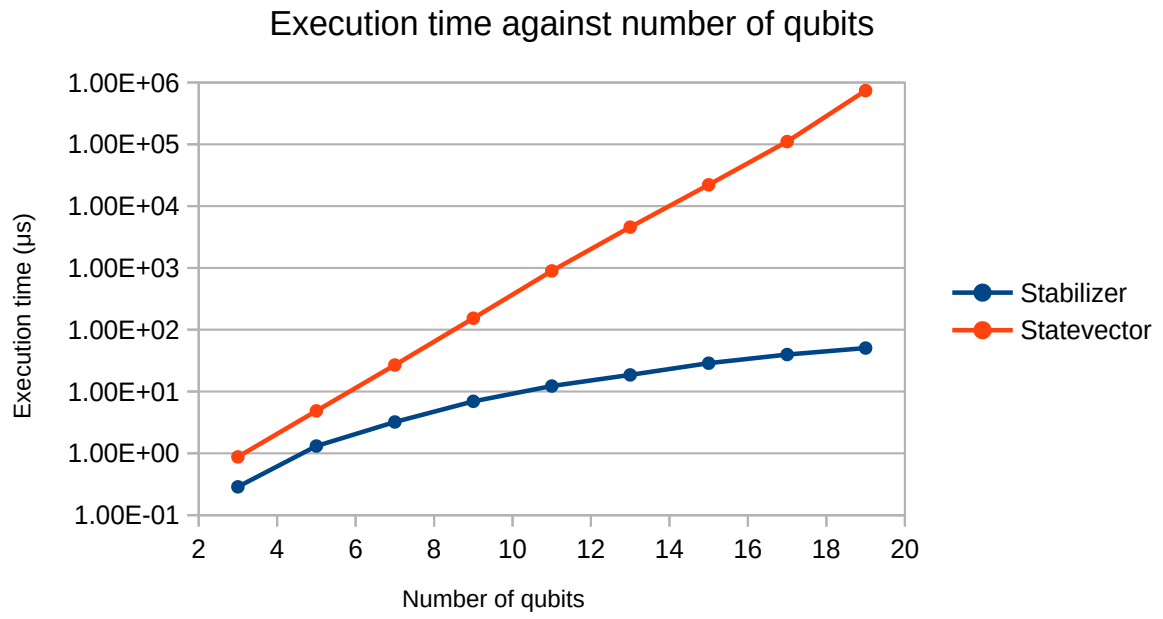
This experiment differs in the last in that the circuit size remained constant at $n = 17$, this time the independent variable was the target qubit. Each kernel was aside from AVX was tested at all target qubit positions from zero to $n - 1$. The AVX kernel cannot accept a target qubit equal to $n - 1$ since the target amplitudes are interleaved with a stride equal to one. Running this experiment provided the following results:



The AVX implementation once again, consistently outperforms the other implementations, at every measurement value it can be applied to. There is an evident increase in execution time as the target approaches $n - 1$. However on the final three targets there are peaks at $n - 3$ and $n - 1$ with a trough at $n - 2$. This pattern was reproduced at various constant qubit counts.

4.4 Stabiliser simulator

The following experiments evaluate the performance of the full statevector and stabiliser simulation objects when provided with the same circuit. The circuit consists of a Hadamard gate with a target qubit one and a CNOT gate with a target qubit zero and control qubit one.



A graph

5 Conclusion

Conclusion

This paper investigated the computational challenges of classical statevector quantum circuit simulation and evaluated strategies to mitigate its inherent bottlenecks at both the algorithmic and hardware levels. By developing a custom Rust-based simulator, this work demonstrated that directly operating on amplitude pairs bypasses the prohibitive memory and computational overhead of full-system matrix expansion.

The experimental results confirmed that Direct Pair Multiplication fundamentally alters memory and execution scaling. At 13 qubits, direct indexing achieved speedups of 44,391.94 times for single-qubit Hadamard gates and 147,010.09 times for controlled two-qubit CNOT gates. While full-system matrix construction required up to 1.34 GB for single-qubit operations and 2.42 GB for controlled two-qubit operations at 13 qubits, direct indexing operated without any additional allocations beyond the amplitude buffer itself.

Furthermore, hardware-accelerated kernels provided substantial execution time reductions over the portable implementation. The AVX SIMD implementation delivered an additional 158% to 225% speedup over the portable kernel. However, memory allocation profiling revealed that performance decreases as the target qubit approaches $n - 1$.

Future research directions could consist of providing measurement operations to the stabiliser simulation for running consequential quantum circuits, as well as investigating the benefits of multithreaded strategies on execution time.

Ultimately, while classical statevector simulation remains bound by exponential state space scaling, combining direct pair multiplication with SIMD vectorization significantly improves runtime and memory efficiency. These optimizations provide a fast, memory-lean environment for designing and verifying quantum algorithms on classical hardware.

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Appendix

Qubits	Full-system matrix (ms)	Direct indexing (ms)
3	0.000623	0.000033
5	0.009367	0.000109
7	0.15016	0.000343
9	2.469403	0.001218
11	47.624858	0.004629
13	793.488575	0.017875

Table 1: Hadamard gate execution time for full-system matrix expansion and direct indexing.

Qubits	Full-system matrix (ms)	Direct indexing (ms)
3	0.001106	0.000036
5	0.016174	0.000075
7	0.245583	0.000188
9	5.016699	0.000642
11	94.779459	0.002286
13	1366.488455	0.009295

Table 2: CNOT gate execution time for full-system matrix expansion and direct indexing.

Qubits	Hadamard (MB)	CNOT (MB)
3	0.001344	0.002496
5	0.020544	0.037056
7	0.327744	0.590016
9	5.242944	9.437376
11	83.886144	150.995136
13	1342.177344	2415.919296

Table 3: Peak memory usage of full-system matrix expansion for Hadamard and CNOT gates.

Qubits	Portable (μ s)	FMA (μ s)	AVX (μ s)
3	0.032974	0.029392	0.018314
5	0.107920	0.073119	0.047978
7	0.352090	0.271980	0.164130
9	1.205300	0.918370	0.622740
11	4.576000	3.254700	2.513300
13	18.508000	12.715000	10.356000
15	72.479000	49.887000	41.201000
17	280.930000	202.460000	163.710000
19	1203.100000	857.450000	760.220000

Table 4: Hadamard gate execution time for hardware-accelerated kernels.